

Visualizing and Predicting Traffic Congestion Rates

Team 3 (Cudly Coders)

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1 INTRODUCTION

Traffic congestion is a major problem in cities like Atlanta, leading to wasted time, higher fuel use, and frustration for drivers. Our project focuses on forecasting and visualizing traffic volume using real-time data from the GDOT and hourly weather data from Weather Underground. Since weather affects how people drive, combining the two can reveal useful traffic patterns. We used four approaches, SARIMAX, LSTM, random forest, and a diffusion-based model, to compare how well different methods handle short-term traffic prediction. The final predictions are shown on a map-based interactive dashboard that displays both traffic and weather trends, helping users and planners anticipate congestion.

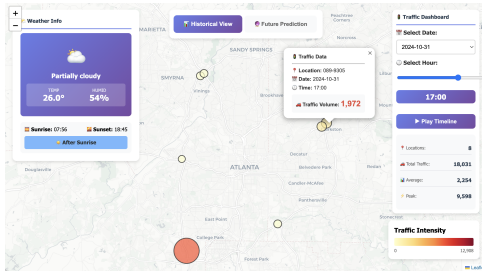


Figure 1: Final Version of the Visualization Web Application

2 PROBLEM DEFINITION

Our project aims to predict short-term traffic volume in Atlanta by combining GDOT traffic sensor data with hourly weather information. Each GDOT station records how many vehicles pass per hour, while weather data add factors such as temperature and rainfall that influence driving patterns.

We model the problem as a time series prediction task, in which we aim to use past traffic data and recent weather conditions to estimate future traffic volume for the next hour at each GDOT station. The models look for patterns over time and predict vehicle congestion through a site in the next hour.

3 LITERATURE SURVEY

Traffic Visualization and Congestion Analysis

Main Ideas: The following studies explore methods for visualizing and quantifying large-scale traffic patterns. Clarinval and Dumas review city visualization techniques such as congestion monitoring and mobility pattern exploration [3]. Seong et al. proposes space-time congestion metrics like weighted and normalized indices [16]. Lee et al. and Chen et al. combine forecasting with interactive dashboards [10] [1], while Wang et al [19]. visualize trajectory data using graph structures.

Why It Is Useful: These papers guide how to transform vehicle data into interpretable visualizations. Their metrics and visualization methods inform our plan to map congestion integrate both observed and predicted data into a visualization.

Shortcomings: Most rely on aggregated sensor data rather than video-level information. They lack generative or predictive modeling, which our project addresses through diffusion forecasting.

Diffusion and Time-Series Models for Spatio-Temporal Traffic Forecasting

Main Ideas Diffusion models have emerged as powerful tools for spatio-temporal prediction. Yang et al. survey diffusion models by task and domain [20], while Sheng et al. introduce NPDiff to improve temporal forecasting via structured noise priors [17]. Yu et al., Kashyap et al., Kar et al., and Lv et al. examine deep learning models such as ST-GCN, CNN, RNN, and YOLO for traffic flow and edge-based estimation [21] [7] [6] [12]. More recent works also explore diffusion models more directly for systems related to transportation. Peng et al. [15] give an overview of how diffusion models, first made for image generation, are sometimes used for transportation system forecasting and data filling now. Chi et al. [2] shows how traffic flow can be turned into "traffic images", which a diffusion model learns to clean and use to make future congestion maps. Lin et al. [11] propose SpecSTG, a faster version of diffusion that works in spectral space, which makes real-time

predictions easier. Zhang et al. [22] take this further and separate high and low frequency patterns to handle uncertainty in complex traffic data. Time-series models are still widely used for predicting traffic flow and volume. He et al. [4] uses an SARIMAX model to forecast short-term traffic by learning the regular and seasonal trends. Kumar and Vanajakshi [9] build on this similar principle using a seasonal SARIMAX version that captures repeating daily and weekly cycles. Katambire et al. [8] compare SARIMAX and LSTM and find that SARIMAX work well when patterns are steadier, while LSTM performs better when the data has noise or is unpredictable. These approaches gave us a good starting point to compare older methods with new diffusion-based models in our project.

Why It Is Useful: These studies form our methodological base. Diffusion models suggest ways to denoise and predict temporal traffic patterns, while deep networks provide benchmarks for spatio-temporal learning. Together, they enable generative congestion forecasting with interpretable spatial dependencies.

Shortcomings: Most prior work focuses on network or sensor data, not video-based scenes, and lacks visual analytics. We extend these ideas by applying diffusion modeling directly to traffic density maps.

4 PROPOSED METHOD

Intuition While diffusion models were initially intended for image generation, they have also shown consistently better performance in non-image applications than deterministic neural nets where noise is high and distribution is complex. Diffusion based models are fundamentally different than conventional forecasting models because they are used to model full conditional distribution of future traffic instead of giving a single deterministic prediction. As one might expect, traffic is inherently noisy, so trying to predict a single exact value (point prediction) usually fails in the most important cases. In our scenario, this means we are not just modeling a single statistic but rather modeling uncertainty, which allows us to successfully model rare events like traffic spikes and sudden drops in congestion. Equally important to highlight, using a diffusion model built on top of an LSTM works better than using one built on a Transformer for this problem. This is due to short-term traffic patterns, the behavior of individual road segments, and daily/weekly cycles being

smooth and sequential in nature. LSTMs are good at modeling these kinds of continuous time-series scenarios. Additionally, when input sequences are long and the training needs to be stable, LSTMs tend to perform better than transformers. The LSTM-backed diffusion model denoises while perserving true temporal structure, enabling it to learn correlation such as gradual congestion buildup, smooth post-jam recovery, and dependencies across adjacent time steps. On days with heavier traffic, diffusion models are more generalizable since they are robust to non-stationarity. Generating multiple samples allows the model to estimate confidence intervals, the probability of congestion within an hour, and worst case travel times more reliably than traditional traffic predictors.

Data

Data Collection Our data comes from the Georgia Department of Transportation (GDOT) Traffic Analysis & Data Application (TADA), which provides detailed traffic volume data for specific highway stations in Georgia [14]. We focused on 10+ sites located around Atlanta’s I-285 perimeter, since it is one of the most traffic-heavy regions in the state. Each site provided separate Excel files with hourly vehicle counts recorded over January 2023 to Mid-October 2025 in 5-week intervals.

We downloaded roughly 400 to 480 Excel files manually, each containing slightly different column formats and missing values. Using Python and Pandas, we cleaned and merged all of the datasets into one consolidated CSV file, which contained over 260,000 data points. This merging process required standardizing the column names, aligning timestamps, and assigning each data entry a unique site identifier (so that the model could see regional differences in flow).

We used hourly weather data from Weather Underground [18] to get weather metrics like humidity, precipitation, and cloud coverage using webscraping [13]. With this, we were able to match each weather entry to the corresponding timestamp in the GDOT data. Since weather patterns influence driving behavior, we treated weather as a variable that adds context to traffic volume patterns. We also performed several preprocessing steps. For example, a necessary step was to convert all timestamps to a consistent format and ensure we had

chronological order in the sequences. Then we normalized some numeric values for stability during model training. Finally, we also performed data imputation for the weather data columns. For the lag features that we created, we dropped rows with null values; these rows were dropped as they were included in the next row. Once we had a cleaned dataset, we split the data into training and testing sets.

Preprocessing and Feature Engineering Following data collection and merging, we performed several preprocessing and feature engineering steps to prepare the dataset for time-series modeling. First, as noted, the `date_time` column was standardized to a consistent `datetime` object.

From this temporal column, we extracted cyclical features known to influence traffic patterns:

- **hour**: The hour of the day (0-23).
- **dayofweek**: The day of the week (0-6).
- **month** and **dayofyear**: Features to capture broader seasonal trends.

In the diffusion model, we also mapped each `siteid` to an integer starting from 1.

To provide the model with explicit historical context, we created several lag features based on the volume variable. These features are critical for time-series forecasting as they allow the model to learn from past patterns:

- **volume_lag1h**: The traffic volume from 1 hour prior.
- **volume_lag24h**: The volume from 24 hours prior, to capture daily cyclical patterns.
- **volume_lag168h**: The volume from 168 hours (1 week) prior, to capture strong weekly seasonality.

Categorical features, such as the `Wind` column, were converted into a numerical format using one-hot encoding (e.g., `WindDir_SW`, `WindDir_North`). Standard Scalar was also applied to features in the diffusion model. As mentioned in the dataset collection, rows containing null values, which were primarily generated at the beginning of the dataset from the creation of these lag features, were dropped prior to model training.

Models

Most current traffic prediction systems show traffic status but cannot anticipate sudden changes. Our method

uses time-series models and diffusion to predict and visualize traffic volume in the near future, specifically for the next hour. While our implementation ultimately uses diffusion, we used SARIMAX, LSTM, and Random Forest as baseline models to compare performance and confirm that diffusion is the most appropriate methodology for our problem.

Baseline Models SARIMAX captures general seasonal patterns, LSTM learns more complex time dependencies, and Random Forest models nonlinear relationships and interactions temporal features. However, Random Forest tends to smooth out extreme peaks. Diffusion models differ because they explicitly handle irregular, noisy changes in traffic by modeling any uncertainty. By comparing all four approaches, we can assess not only the accuracy of each method but also their ability to adapt to the unpredictability of traffic and weather conditions.

The LSTM model had one recurrent layer with 50 units (using ReLU) and a final Dense layer for the output. It took a 24-hour rolling input window with eight features, including lagged traffic volumes (1h, 24h, 168h), time-of-day and day-of-week indicators, a site ID embedding, and two weather variables (solar radiation and temperature). We trained the model on data from GDOT site 063-1201 using MinMax scaling and an 80/20 chronological train-test split. Training used 10 epochs, a batch size of 32, and a 10% validation split. On the test set, the model achieved an RMSE of 667 and an MAE of 464.

For the ARIMAX model, we used the station with the longest historical record. We converted its hourly data into daily average traffic volumes and matched each day with one weather reading taken at midnight (or the earliest available time). We then fit a SARIMAX model with order (1,1,1) and seasonal order (1,1,1,7), using the daily weather variables as external predictors. The model was estimated with maximum likelihood, and we allowed the stationarity and invertibility checks to be relaxed so the model could fit the data more easily. Performance was evaluated with an 80/20 chronological split, and forecasts were produced by providing the model with the corresponding daily weather inputs.

The Random Forest model was trained on data from site 063-1201 using an 80/20 time-ordered train/test split. It used seven input features: three lagged traffic volumes

(1h, 24h, 168h), two temporal indicators (Hour, Day-Of-Week), and two weather variables (solar radiation and temperature). The model was configured with 100 trees and a maximum depth of 15. Among the baseline models we evaluated, this setup produced the best overall performance.

For our analysis and modeling, we used several Python libraries. These included Pandas for data handling, Statsmodels for the SARIMAX model, Matplotlib for plotting, TensorFlow/Keras for the LSTM (using Sequential, LSTM, and Dense layers), NumPy for numerical work, and scikit-learn for the Random Forest model, model selection tools, and performance metrics.

Diffusion In order to format the data for the LSTM-based diffusion model, we created 24-hour windows of traffic data using pandas, which made it easier to handle the temporal nature of the data as well as separate it into historical and target data (the final hour in the window). Then, noise was added using a pre-established, tuned noise schedule within the forward diffusion process.

The resultant noisy data was then the input of a single-layer LSTM encoder, built using torch and chosen for its ability to handle time-series data. The encoded output was denoised using time embedding, which converted the discrete times into a more stable continuous sinusoidal representation, and a standard-practice two-layer perceptron that includes sigmoid linear unit and batch normalization, resulting in a one-dimensional output representing the target hour’s prediction [5]. An 80-20 split was used in the `train_test_split` method (from numpy) to create our training and testing data, and was then converted into compatible `TensorDataset` and `DataLoader` formats. Training was conducted for 10 epochs using the Adam optimizer, and predictions were generated on the remaining data.

Visualization

We developed an interactive web-based dashboard using Python to generate a self-contained HTML application. The visualization system consists of three main components: data compression, visual encoding, and interactive features.

Data Compression Strategy Given the substantial size of traffic and weather datasets (over 330,000 rows combined), efficient data compression was critical for

creating a deployable single-file application. We implemented a compression pipeline involving structural optimization, Gzip compression, and Base64 encoding. First, raw csv data was transformed into compact JSON arrays, replacing verbose key-value pairs with positional array, reducing payload size by approximately 40%. Then, the optimized JSON structures were compressed using Python’s gzip library with maximum compression level, achieving an additional 60 – 70% reduction. Finally, compressed binary data was encoded to base64 for safe embedding within HTML, making the entire application portable as a single file. Overall, we achieve a compression ratio of approximately 75%, reducing the final HTML file size to under 2MB while maintaining the fidelity of the full dataset.

Visual Encoding Design To intuitively represent traffic volume, we employed both circle size and color gradient. Traffic volume is mapped to marker radius in pixels: $r = 8 + 32(\frac{\text{traffic volume}}{\text{max traffic}}) \in [8, 40]$. An 8-step color scale represents traffic density through proportional binning. This leverages pre-attentive visual processing, allowing users to quickly identify both absolute and relative volumes across the entire traffic network at a glance.

Interactive Features While aggregate comparison across sites is valuable, users often need to analyze temporal patterns for individual locations. To enable this, we implemented: site-level detail popups upon click events, displaying location identifier, timestamp, traffic volume, and historical vs predicted data context; temporal navigation through data selections, hour sliders, and play/pause animation for timeline progression; and mode switching through historical observation and AI/model prediction views while maintaining consistent visual encoding for easy comparison

Technology Stack The dashboard is built on `Leaflet.js` for geospatial visualization, `Pako.js` for client-side decompression, Javascript for portability, and responsive CSS with media queries for cross-device compatibility. This enables both quick pattern recognition through visual encoding and detailed investigation through interactive exploration, supporting diverse analytical workflows for traffic prediction analysis.

5 EVALUATION

Experimental Setup

Testbed Data was collected from GDOT and Weather Underground via web scraping and manual downloads, and preprocessed to interpolate missing values, create new features, scale values, and create sliding windows. SARIMAX, Random Forest Regressor, and LSTM were used as baseline models to compare against the performance of the Diffusion-LSTM model. A map-style visualization was created using JavaScript and CSS. Further implementation details can be found in Section 4.

Questions Our experiments were designed to answer the following key questions:

Accuracy: How does LSTM-based diffusion compare to more traditional models when predicting short-term traffic congestion? Is it more robust to noise?

Feature Importance: How meaningful are the features in predicting traffic congestion for the next hour?

Computational Efficiency: Is the performance gain worth added time complexity?

Traffic: Can short-term traffic predictions make meaningful impact on the understanding of traffic buildup to improve city-planning?

Observations

Table 1: Model Performance Comparison

Model	RMSE	MAE
Random Forest	561.9	361.0
LSTM	667.3	464.8
SARIMAX	1348.6	1085.0
Diffusion	510.4	255.3

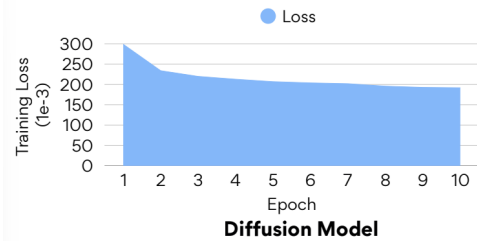


Figure 2: Loss During Training

Model Performance As summarized in Table 1, the Random Forest Regressor achieved the strongest overall performance among the baseline models, with the

lowest RMSE and MAE compared to LSTM and SARIMAX. The diffusion model outperformed all baseline approaches with the lowest error in context of true mean of 1539.5 and an R^2 of 0.95. The diffusion training trajectory showed a smooth and stable reduction in loss in Figure 2, indicating consistent learning performance.

Baseline Context Random Forest outperforming the other baseline models indicates that our data might be more driven by instantaneous features or nonlinear interactions than by long sequential memory. This informed our decision to use 24-hour time windows over longer windows for our diffusion model. Additionally, the underperformance of the other two indicates that there may be quite a bit of noise, something diffusion is uniquely equipped to handle.

Error Behavior Key findings from error analysis include:

- Random Forest followed local fluctuations most accurately but still smoothed sharp spikes.
- SARIMAX struggled with nonlinear patterns and abrupt changes.
- LSTM captured long-term weekly structure but often failed on sudden shifts.

The Diffusion Model, in contrast:

- Modeled uncertainty explicitly.
- Represented multi-modal outcomes (multiple plausible futures).
- Handled unusual dips and spikes more robustly.

Overall, the diffusion model combines the strengths of our baseline models, letting us capture subtle, long-term trends while still emphasizing the relatively instantaneous nature of short-term traffic prediction.

Feature Insights Random Forest feature importance analysis revealed that:

- Hour-of-day and day-of-week were the strongest predictors.
- Lag features (1-hour, 24-hour, 168-hour) provided substantial predictive power.
- Weather features contributed minimally, suggesting that traffic flow is primarily driven by temporal cycles rather than meteorological factors.

This supports the observation that traffic behavior is highly cyclical and only weakly influenced by traditional weather occurrences, and helps to explain why Random Forest performs relatively well.

Model Insights

- **Random Forest:** Performed the best out of the baseline models. Handled most normal traffic patterns well. Still struggled with big changes or sudden changes.
- **SARIMAX:** Easy to interpret but did not handle Atlanta’s irregularities in traffic well. Worked fine for smooth trends but not for irregularities.
- **LSTM:** Learned the overall weekly and daily patterns in the data. Needed more tuning and training. Predictions were smoother, so it sometimes missed sudden changes.
- **Diffusion Model:** Handled randomness and noise better than the other models. Made the most accurate predictions one hour ahead. It was the best at dealing with messy traffic data.

Visualization We developed a fully interactive, single-file web dashboard to evaluate model outputs and explore traffic dynamics across the I-285 region.

- Visualization of over **267K historical** and **53K prediction** traffic records revealed clear daily and weekly patterns as well as irregular spikes
- Marker color (yellow to red) and size encoded traffic intensity effectively highlighted areas with consistently high congestion, allowing rapid identification of bottlenecks for the user.
- Geospatial rendering using Leaflet.js made it easy to detect clusters of high traffic and regional flow pattern
- Hourly weather integration and sunrise/sunset for historical mode showed the subtle but limited effect of weather on traffic
- Prediction mode with date/hour controls synchronized to output.csv to show the performance of our model in comparison to ground truth data, allowing us to see where our model underperformed
- Sunrise/sunset extraction for 1,027 dates allowed observation of potential traffic-lighting patterns and changes in traffic behavior
- Data compressed and client-side decompression produced fast load times (1-3 seconds) to enable smooth exploration of a large dataset
- Fully self-contained HTML bundle for easy sharing and deployment facilitated sharing insights without additional setup

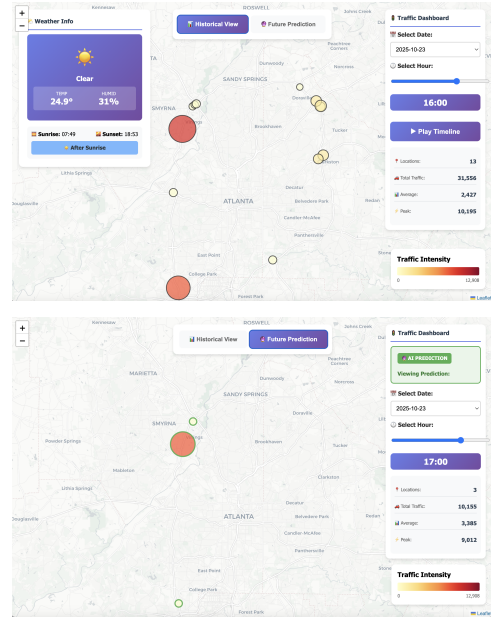


Figure 3: Visualization Web Application. The top panel shows the historical view, and the bottom panel shows the future prediction.

6 CONCLUSIONS AND DISCUSSION

Through experiments with SARIMAX, LSTM, and Random Forest Regressor, we found that traditional baseline models struggle with irregular or noisy traffic patterns, and weather contributes minimally compared to temporal features such as traffic volume and day of the week. The diffusion model demonstrated the most accurate and stable short-term forecasting performance, making it well-suited for real-time travel decisions such as route selection and congestion avoidance.

While the current system predicts traffic one hour ahead, results indicate strong potential for scaling to multi-hour (12–24 hour) forecasting with further training. Additional extensions should include accounting for spatial relations as well, especially on roadways like the Atlanta perimeter.

This foundation supports future development toward real-time traveler guidance as well as broader city-level traffic management and planning. The potential benefit for a full-scale deployment extends to daily commuters, city planners, and fleet operators.

All team members have contributed a similar amount of effort.

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